

In Ethereum, there are two types of addresses: the *externally owned account* (EOA) and an address of a *contract account*. Transactions sent to an EOA can only transfer ETH.¹ In Bitcoin, all addresses are EOA. In Ethereum, when data is sent to a contract account, the data are used to execute code in that contract. The transaction may or may not have an accompanying ETH payment for use by the contract.

A single transaction starts with an end user from an EOA but can interact with many dApps (or any Ethereum smart contract) before completing. The transaction starts by interacting with a single contract, which will enumerate all the intermediate steps in the transaction required within the contract body.

Clauses in a smart contract can cause a transaction to fail and thereby revert all previous steps of the transaction; as a result, transactions are *atomic*. Atomicity is a critical feature of transactions because funds can move between many contracts (i.e., exchange hands) with the knowledge and security that if one of the conditions is not met, the contract terms reset as if the money never left the starting point.

Remember that transactions have a gas fee, which varies based on the complexity of the transaction. When, for example, ETH is used to compensate a miner for including and executing a transaction, the gas fee is relatively low. Longer or more data-intensive transactions cost more gas. If a transaction reverts for any reason, or runs out of gas, the sender forfeits all gas used until that point. Forfeiture protects the miners who, without this provision, could fall prey